

E-Cell AI & Automation - Task 3

Customer Intelligence, Agents, Cohort Analysis & HEART Metrics

- AI-Integrated CRM Platform for E-Cell Operations
- Local LLM inference via Ollama (Llama-3-8B-Instruct)
- GitHub: [KanakMalpani/Ecell-Projects](#)

Problem Statement

- Startups lack AI-native CRM tools for tickets, interactions, and workflows
- Build enterprise-grade CRM with LLM summarization and agent automation
- Measure impact via HEART framework and cohort retention analytics

Five-Module Architecture

- Module 1: Customer & ticket CRUD, segmentation, timeline views
- Module 2: LangChain summarization + LangGraph agent workflows + memory
- Module 3: Cohort retention curves, churn scoring, JSON/PDF export
- Module 4: HEART dashboard (Happiness, Engagement, Adoption, Retention, Task Success)
- Module 5: FastAPI backend with RBAC and audit metadata

Synthetic Dataset

- 520 customer profiles across 6 industries and 3 product tiers
- 1,050+ support tickets with full lifecycle states
- 2,500+ interaction logs spanning ~6 months
- Bulk ingestion with validation and deduplication on startup

LLM Stack (Ollama Local)

- Ticket summarization: key issues, urgency, resolution path
- LangGraph pipeline: load context -> route -> generate -> escalation
- Hallucination guard: source citations, confidence scoring, flag unsourced claims
- Per-customer short-term + long-term interaction memory

Role-Based Access Control

- Agent: create/query tickets, run summarization and agent pipeline
- Supervisor: agent permissions + cohort analysis read access
- Admin: full system access across all CRM modules
- Analytics: read-only HEART metrics, cohorts, and customer data

Core API Endpoints

- POST /api/v1/customers - create customer with cohort assignment
- POST /api/v1/tickets/create - ticket lifecycle management
- POST /api/v1/tickets/{id}/summarize - LangChain summarization
- POST /api/v1/query/agent - LangGraph multi-turn agent
- GET /api/v1/cohorts/analysis - retention curves and churn scores

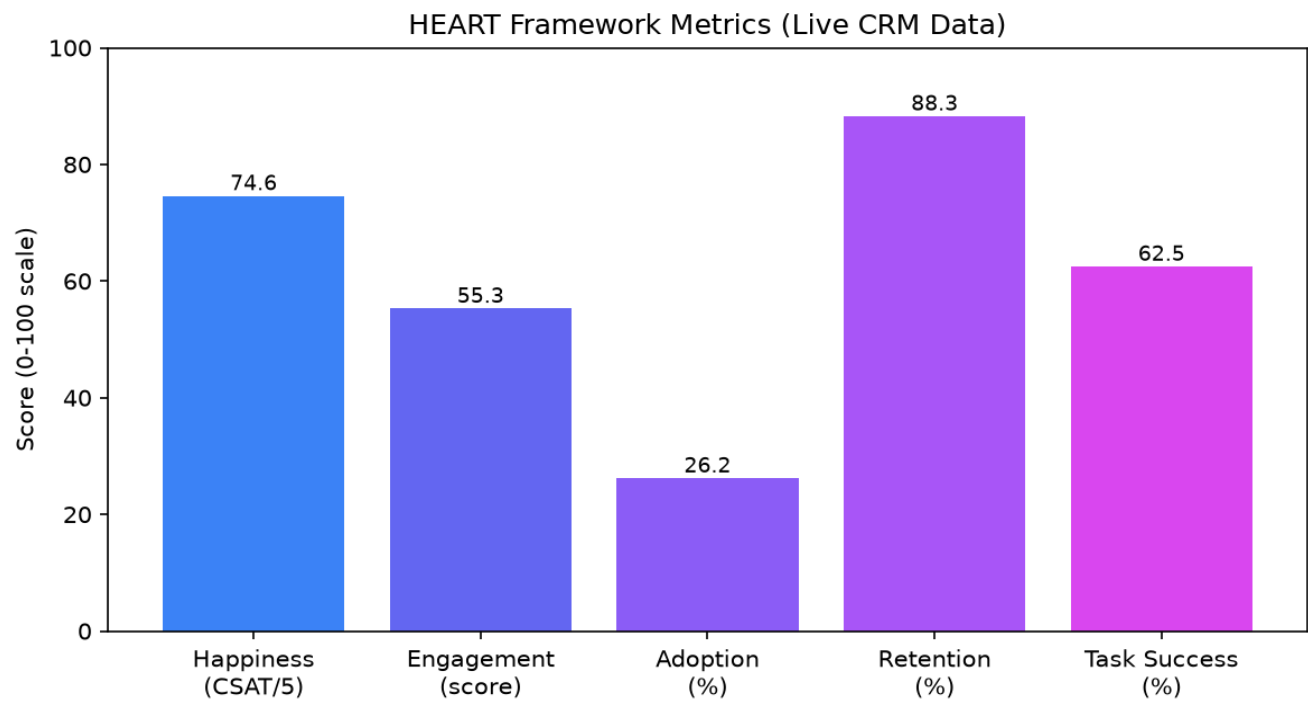
HEART Framework Metrics

- Happiness: CSAT and NPS proxy by cohort/channel
- Engagement: active customers, ticket open rates, session depth
- Adoption: AI-assisted ticket rate, onboarding completion
- Retention: monthly retention, churn flags, lifespan by segment
- Task Success: resolution rate, FCR, escalation frequency

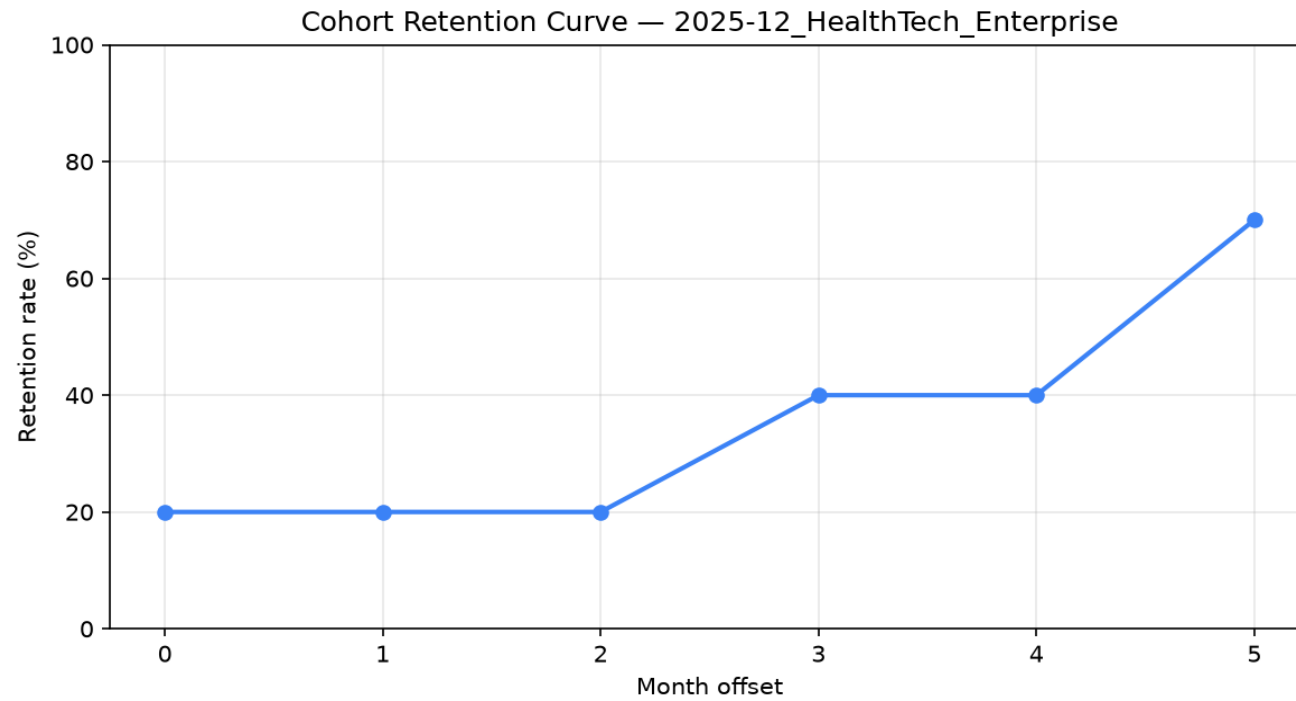
Evaluation Results (Live CRM Data)

Metric	Value
Avg CSAT (Happiness)	3.73 / 5
Active customers (Engagement)	430
AI-assisted ticket rate (Adoption)	39.0%
Monthly retention (Retention)	88.3%
Ticket resolution rate (Task Success)	62.5%
Churn model F1	1.00
Churn-flagged customers	130
Largest cohort M+6 retention	70.0%

HEART Metrics Chart



Retention Curve - 2025-12_HealthTech_Enterprise



Live Demo

- Swagger UI: <http://127.0.0.1:8002/docs>
- HEART dashboard: <http://127.0.0.1:8002/dashboard>
- Agent query with memory-aware multi-turn context
- Cohort report export to JSON and PDF

Deliverables

- Source code: AI and Automation/Task-3/ on GitHub
- System report: SYSTEM_REPORT.md + submission/SYSTEM_REPORT.pdf
- Presentation: submission/Task-3-Presentation.pdf
- Run: `python run_pipeline.py` then `uvicorn api.app:app --port 8002`